Speech Enhancement versus Speech Recognition

A multi-speaker study of STFT Wiener filtering with real environmental noise

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Abstract

This project investigates whether waveform improvement after STFT Wiener filtering is accompanied by better intelligibility prediction and automatic speech recognition. A development set of four speakers selects one gain floor from 0.05, 0.10, 0.20 and 0.35. A disjoint test set contains 20 utterances from ten speakers, equally divided by recorded sex. White noise and real DEMAND traffic/cafeteria recordings are mixed at -5, 0, 5 and 10 dB, giving 240 test mixtures. Fixed Whisper tiny.en inference, SNR and STOI evaluate noisy, baseline, selected-floor and oracle-PSD conditions. The selected floor is 0.35. Baseline mean SNR changes from 2.50 to 7.24 dB, while corpus WER changes from 28.28% to 29.90%. Selected-floor WER is 25.15%. There are 69 baseline cases in which SNR rises but WER worsens. The results establish a task-dependent trade-off within this controlled subset, without claiming general superiority across recognisers or acoustic environments.

1. Introduction and research question

Question: Does STFT Wiener filtering improve SNR, STOI and recognition consistently on previously unseen speakers, and can a gain floor selected on separate development speech improve the recognition trade-off?

A denoiser may remove speech along with interference. Samplewise error reduction therefore does not necessarily preserve the acoustic distinctions needed for word recognition. Enhancement objectives can differ [1], so this study measures waveform fidelity, predicted intelligibility and machine recognition on exactly matched outputs. It tests a transparent classical method rather than training a large model.

The completed work includes a reproducible filtering baseline, a small development-selected modification, speaker-disjoint evaluation, recorded environmental noises, speaker-level uncertainty analysis and jointly scaled listening examples. It extends the earlier eight-utterance pilot; the new evaluation replaces that pilot as the main evidence.

Guide: literature p.2; data p.3; DSP and metrics p.4; selection/results pp.5-7; audio analysis p.8; discussion p.9; conclusion/references p.10.

Project: GitHub Pages | code repository | companion video | audio examples. Addresses inherited from the project materials; publishing the revised files remains a submission step.

2. Literature review and approach selection

2.1 Spectral subtraction and Wiener estimation

Boll [2] subtracts an estimated background spectrum from noisy speech and reconstructs a waveform. Its appeal is a simple, processor-independent front end, but reliable noise estimation is central to its operation. The present pipeline uses nonnegative power subtraction inside a Wiener-style gain; it is not an exact reproduction of Boll's magnitude-subtraction implementation. This distinction matters because similar-looking spectral operations optimise different quantities and can introduce different distortions.

Lim and Oppenheim [1] review enhancement under different objectives and constraints. For uncorrelated additive speech and noise, minimising expected squared error in one spectral bin yields a power-ratio Wiener gain. Here the true powers are unknown and are replaced by finite-sample estimates. That plug-in approximation is computationally convenient but lacks a guarantee of minimum error on each utterance. A lower gain bound additionally trades interference attenuation against speech preservation. These properties make the method suitable for an interpretable parameter experiment.

2.2 Why several metrics are necessary

Hu and Loizou [3] evaluate objective quality measures against separate subjective judgements of speech distortion, noise distortion and overall quality. This warns against interpreting a single waveform number as complete evidence of enhancement. STOI [4] instead predicts intelligibility from paired clean/degraded speech using short-time representations. It is useful without a listening panel, but remains a prediction rather than a measured percentage of words understood. Neither source establishes a universal mapping from SNR or STOI to machine WER.

Whisper [7] supplies a pretrained recognition model that can be held fixed while the front end changes. The tiny.en variant keeps CPU experiments manageable. Its errors depend on the acoustic input, learned language patterns and decoder settings. Consequently, WER is a downstream outcome of this specific recogniser rather than an intrinsic audio-quality measure. Modern DNS evaluations [8] combine recognition-related and perceptual criteria, providing broader motivation for a multi-objective design.

2.3 Data realism and bounded contribution

LibriSpeech [5] supplies transcribed read speech and speaker metadata; DEMAND [6] supplies recorded noise from different environments. Combining them permits controlled SNR while retaining real background variation. It still produces simulated mixtures rather than recordings of speech captured inside those environments. In particular, cafeteria noise may contain competing voices, which a static single-channel filter cannot reliably separate from the target.

Approach	Strength	Limitation for this task
Spectral processing [1,2]	Transparent and light to run	Depends on noise-power estimates
Learned enhancement context [8]	Addresses complex acoustic variation	Requires an additional model and broader evaluation
Fixed pretrained ASR [7,9]	Directly measures recognised words	Results depend on model and decoding

The project contribution is modest: test whether development-based gain-floor selection transfers to disjoint speakers under a fixed protocol. It does not propose a new denoising architecture or claim current leading performance. Background concepts clarified during implementation include power versus magnitude spectra, separate development/test roles, intrusive metrics and the distinction between oracle information and a deployable estimator.

3. Data and experimental design

3.1 Deterministic speech selection

Speech comes from the official LibriSpeech test-clean archive [5]. Within each metadata sex category, numeric speaker IDs are scanned in increasing order. A speaker is eligible if at least two utterances have durations between 3 and 8 seconds; clips are taken in lexicographic ID order. The first two eligible speakers per sex supply one development clip each. The next five per sex supply two test clips each. Selection never uses recognition scores. Both project subsets originate from test-clean; "development" here denotes the project's tuning partition, not the official dev-clean split.

Property	Development	Final test
Utterances	4	20
Speakers	4	10
Speaker overlap	None	None
Recorded sex	2 F / 2 M speakers	5 F / 5 M speakers
Speech duration	24.43 s	108.42 s
Reference words	61	277
Mixtures	48	240

Table 1. Each utterance is tested in three noise types at four SNRs. Repeated mixtures increase condition coverage but not the number of independent speakers. The selected IDs, transcripts, split membership and source-file hashes are included in data/manifest.json.

3.2 Noise recordings and temporal separation

The noise conditions are generated white Gaussian noise and channel 1 of DEMAND STRAFFIC and PCAFETER, at 16 kHz [6]. Development excerpts are restricted to seconds 0-60; test excerpts use seconds 120-300, leaving a 60-second guard. The recordings are shared across project subsets, so the experiment evaluates new speakers and disjoint noise times, not unseen noise environments. Only one recording per real environment is available in this design.

For each mixture, a one-second noise-only calibration interval immediately precedes the noise added to the utterance. These adjacent intervals do not overlap. Both receive the same mixing scale. Calibration is not renormalised to match the subsequent noise power, allowing real level changes to cause estimation error. This improves on the pilot's known-power calibration assumption while still requiring a correctly identified noise-only interval.

3.3 Controlled perturbation and reproducibility

For speech s and noise excerpt v, the input is y = s + a v. The scale a is computed from full-utterance powers, with target L in {-5, 0, 5, 10} dB:

$$a = \mathrm{sqrt}(\mathrm{mean}(s^2) / [\mathrm{mean}(v^2)\ 10^{(L/10)}]).$$

A deterministic SHA-256-derived seed depends on split, utterance ID, noise type and SNR. Every compared method therefore receives exactly the same mixture. Audio remains floating point during processing. One common gain, including the reference and all versions of a mixture, prevents clipping; no method receives an independent loudness normalisation. The achieved SNRs are checked numerically.

4. Signal processing and evaluation

4.1 Analysis, gain and synthesis

All speech and noise are mono at 16 kHz. STFT uses a Hann window of 512 samples (32 ms), a 512-point FFT and a 128-sample hop (8 ms), with zero boundary extension and padding. A time-averaged noise-power estimate P_v(k) is calculated from the one-second calibration interval. Nonnegative subtraction estimates framewise speech power:

$$P_s(k,m) = \max(|Y(k,m)|^2 - P_v(k), 0).$$

For true uncorrelated bin components S and V, $E|S - g(S+V)|^2 = (1-g)^2 P_s + g^2 P_v$. Differentiating with respect to real g gives $g = P_s/(P_s+P_v)$. The implemented plug-in estimate applies a minimum amplitude gain:

$$G(k,m) = \max(g_{min}, P_s(k,m) / [P_s(k,m) + P_v(k) + 10^{-12}]).$$

The output spectrum is GY, retaining mixture phase. Matching inverse STFT reconstructs a waveform, trimmed to input length. A floor of 0.05 permits up to 26.0 dB attenuation; the selected floor of 0.35 permits about 9.1 dB. Raising the floor passes more weak speech and more interference. This is a bounded modification to an existing method, not a new Wiener derivation.

Condition	Noise estimate	Gain floor
Noisy	No filtering	Not applicable
Baseline	Preceding one-second interval	0.05
Selected	Same calibration as baseline	0.35, chosen on development only
Oracle PSD	Exact noise added during the utterance	0.05

Table 2. Oracle PSD is a diagnostic control. It has exact noise-only samples but still averages over time, estimates speech power by subtraction and retains noisy phase. It is not a mathematical upper bound for all metrics.

4.2 Three distinct objectives

SNR: 10 log10(mean(s^2)/mean((s-x)^2)); higher is better. STOI: pystoi with extended=False on aligned reference/output audio; higher predicts better intelligibility. WER: (substitutions + deletions + insertions)/reference words; lower is better. Corpus WER pools errors and reference words. It is not the average of sentence percentages and can exceed 100% through insertions. Text is lowercased, apostrophes removed and alphanumeric words retained.

Whisper tiny.en uses the pinned faster-whisper model revision, CPU int8, English, beam size 1, temperature 0, no prior-text conditioning and no VAD. A fixed three-token n-gram repetition guard and 96-token maximum are applied to every input, including clean and noisy speech. These safeguards limit decoder repetition; they may affect severe-noise WER and are part of the reported configuration. Old-pilot WER is therefore not directly comparable.

Paired differences compare each processed output to its noisy input. A 4,000-draw speaker-cluster bootstrap resamples the ten test speakers while retaining their repeated conditions. Percentile intervals describe variation in this selected speech set conditional on the fixed noise recordings and decoder; no independent-sample significance claim is made.

5. Development selection and final results

5.1 Freeze the parameter before final testing

Each of four candidate floors is applied to the 48 development mixtures. The primary selection criterion is the lowest development corpus WER; a tie prefers the smaller floor. The noisy condition provides context but is not a candidate filter. The selection file is written before the final-test loop starts. No test transcript or score is used to choose the floor.

Candidate floor	Development WER (%)	Errors / reference words
0.05	31.01	227 / 732
0.10	31.69	232 / 732
0.20	31.15	228 / 732
0.35	27.32	200 / 732

Table 3. The chosen floor is 0.35. Four development speakers offer only a small selection sample; the result is an application-specific parameter choice, not a universal optimum. The final test assesses its transfer to new speakers.

5.2 Aggregate final-test performance

Condition	SNR (dB)	STOI	WER (%)	Errors / words
Noisy input	2.50	0.794	28.28	940 / 3324
Baseline (0.05)	7.24	0.798	29.90	994 / 3324
Selected (0.35)	6.56	0.802	25.15	836 / 3324
Oracle PSD (0.05)	7.44	0.800	30.39	1010 / 3324

Table 4. Results over 240 paired test mixtures per method. Clean-reference WER is 5.78% (16/277 words). The noisy-condition denominator repeats the same transcripts across 12 settings and is not that many independent speech observations.

Baseline minus noisy changes mean SNR by +4.74 dB, STOI by +0.004 and WER by +1.62 percentage points. Selected minus baseline changes SNR by -0.68 dB and WER by -4.75 points. These effects concern different objectives and should not be collapsed into a claim that every aspect of audio improved.

Speaker-bootstrap WER-change intervals are [-0.04, +3.00] points for baseline-minus-noisy, [-6.89, -2.51] for selected-minus-baseline, and [-5.01, -1.50] for selected-minus-noisy. Baseline average worsening is uncertain; selected-floor improvements have consistently negative intervals within this fixed-recording design. These are not claims about unseen acoustic environments.

The baseline has 9 outputs above 100% sentence WER; the selected method has 4. These outputs remain in all primary results. Saved edit-operation counts allow errors to be distinguished from waveform quality changes.

6. Results across noise levels and environments

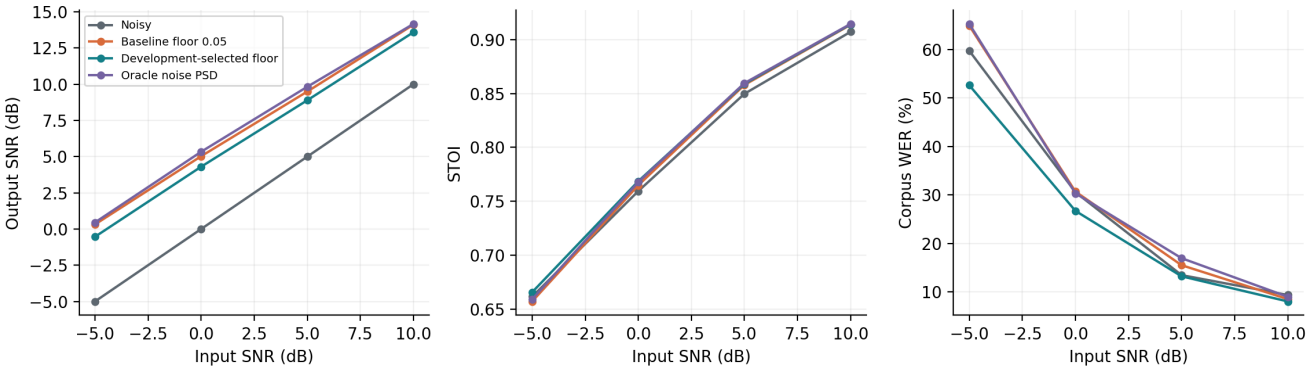


Figure 1. SNR, STOI and pooled WER across input SNRs. Every point uses the same 20 test utterances and three noise types. Lines connect measured conditions.

The three panels answer different questions. SNR measures waveform error, STOI predicts intelligibility and WER measures discrete recognised words. Differences in their trends indicate that a parameter favourable to one target may be less favourable to another. Improvements should be stated relative to an explicit reference condition.

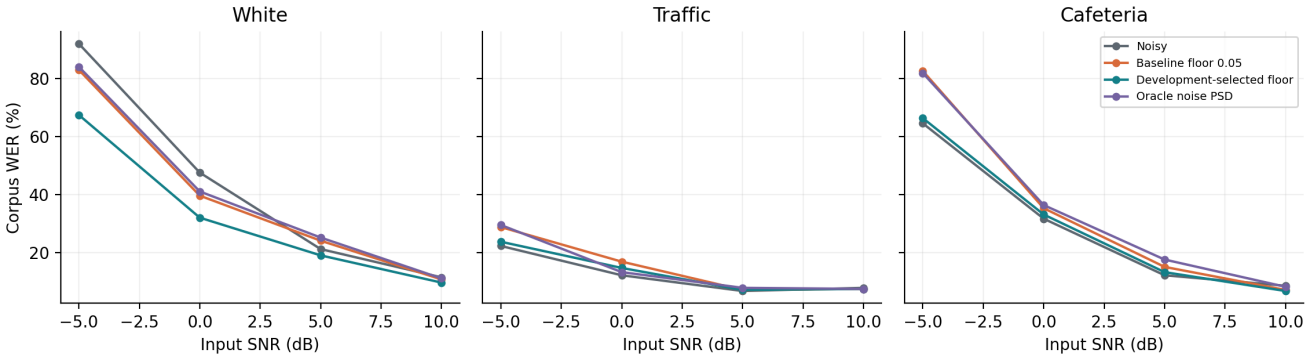


Figure 2. WER separated by noise environment. Each point aggregates 20 speech clips. Traffic and cafeteria are real recordings; white noise is a controlled synthetic reference.

Noise	Noisy WER	Baseline WER	Selected WER
White	43.14%	39.44%	32.13%
Traffic	12.36%	15.16%	13.36%
Cafeteria	29.33%	35.11%	29.96%

Table 5. Noise-wise corpus WER averaged over the four SNR settings through pooled word counts. Equal full-band SNR does not imply equal spectral masking. Real recordings change level and content, so a preceding static calibration can misrepresent the subsequent interference. This mechanism is plausible but is not isolated causally by these curves.

7. Paired outcomes, speakers and word errors

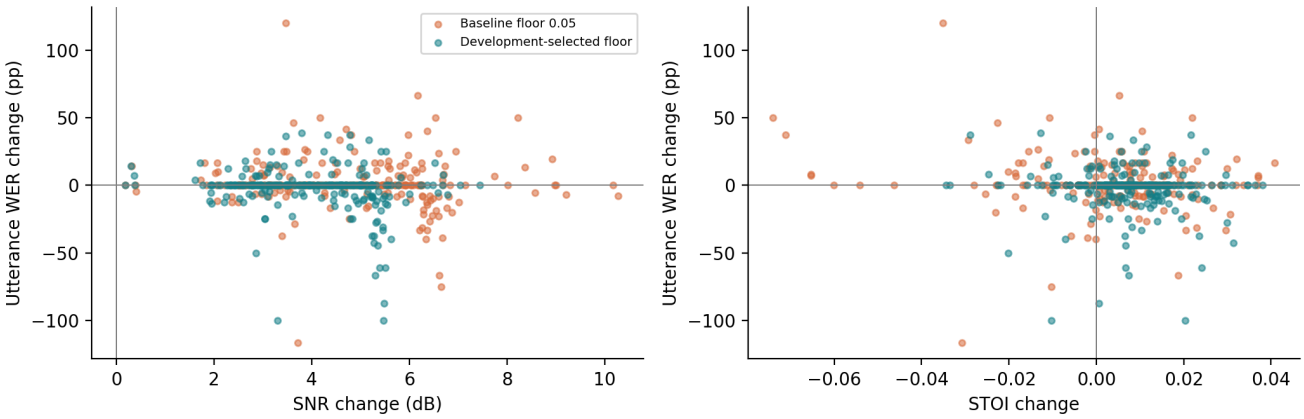


Figure 3. Output-minus-noisy changes for individual test conditions. Positive WER change means worse recognition. Upper-right SNR points show improved waveform fidelity with poorer recognition.

Baseline SNR improves while WER worsens in 69/240 cases; the selected method has 41/240. Baseline SNR improves while STOI falls in 64 cases. Spearman correlation of SNR change with sentence-WER change is -0.127 for baseline and -0.229 for selected. These descriptive correlations do not treat repeated points as independent samples.

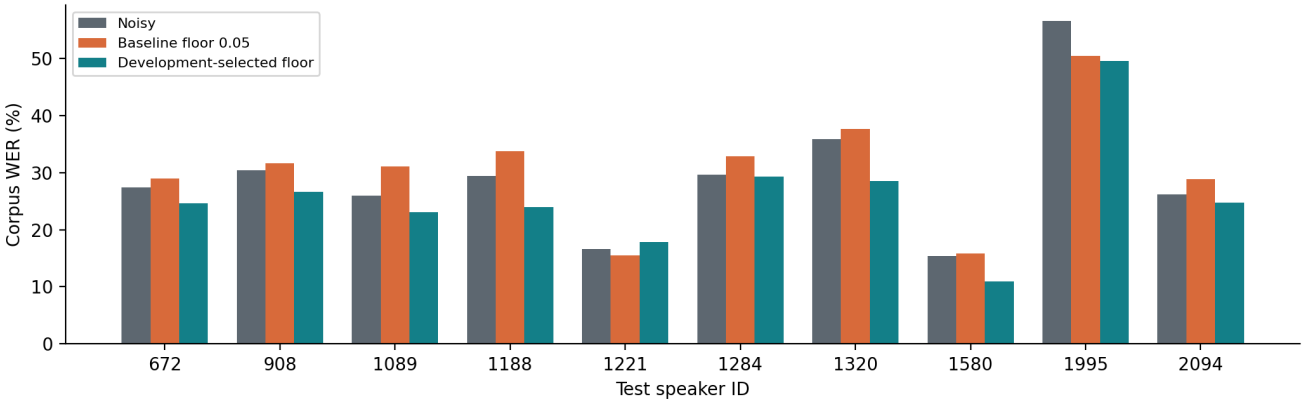


Figure 4. Corpus WER per test speaker across the same noise/SNR grid. Speaker IDs are metadata identifiers, not an ordered measure of difficulty.

The selected method lowers WER against baseline for 9/10 speakers. Between-speaker variation shows why an eight-clip, single-speaker pilot could not establish generalisation. Even this expanded subset is too small to characterise accents, speaking styles or recording conditions broadly.

Condition	Substitutions	Deletions	Insertions
Noisy input	676	139	125
Baseline (0.05)	710	152	132
Selected (0.35)	579	171	86

Table 6. Summed word-edit counts. Removing the same largest baseline error-excess condition from baseline and noisy gives a WER difference of +1.09 points, compared with +1.62 points in the full data. This is sensitivity analysis only; the primary dataset remains unchanged.

8. Signal analysis and listening examples

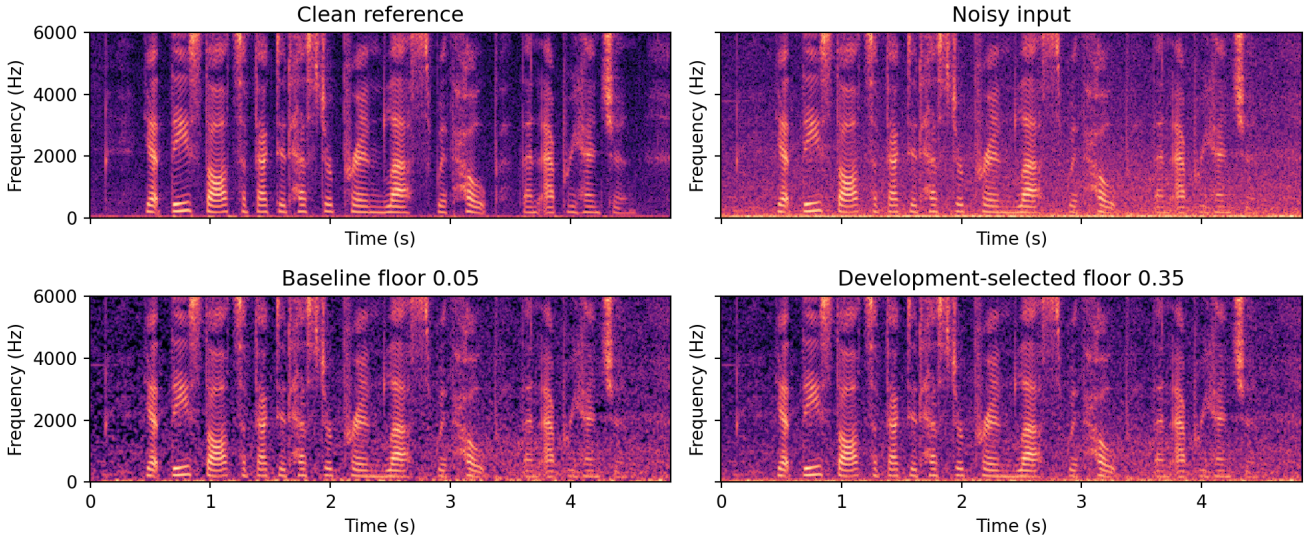


Figure 5. Log-magnitude STFTs for preselected example 1: 1221-135766-0002, traffic noise at 0 dB. Every panel shares the same export gain, frequency range and colour limits. The selected floor is 0.35.

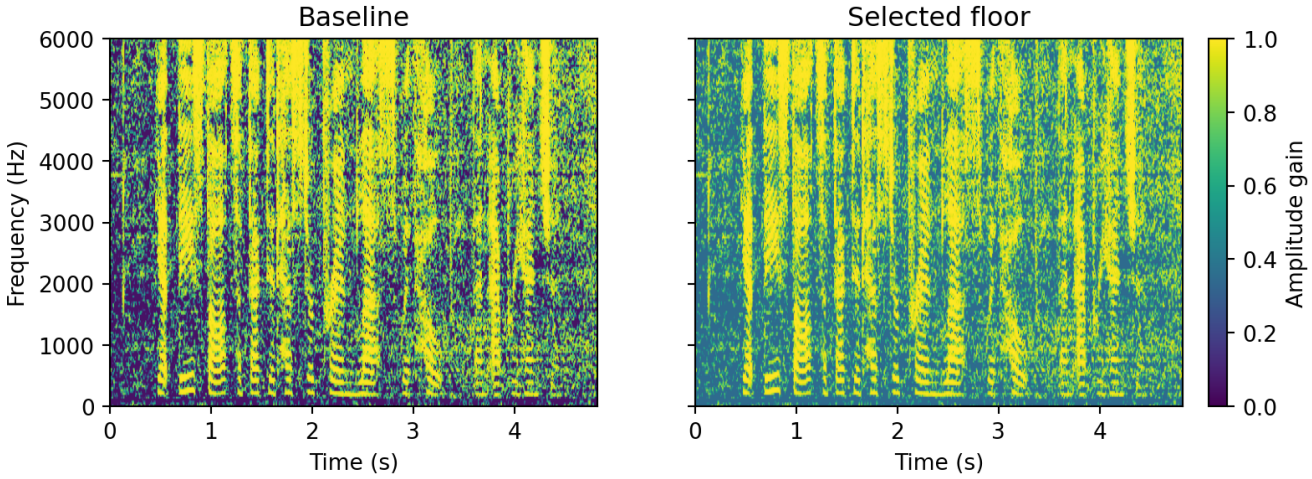


Figure 6. Amplitude gain maps for the same mixture. The modified minimum gain changes strongly attenuated regions. A time-frequency display reveals the applied filter, not whether a particular phoneme survives.

The minimum gain provides a direct mechanism for the trade-off: higher-floor bins retain more mixture energy. This can preserve weak speech components, but also preserves interference. The maps establish how the implementation differs; they do not prove which bins caused a word substitution. No phoneme alignment or model-internal causal analysis is claimed.

Four examples are chosen by the first final-test clip within each sex, crossed with traffic/cafeteria at 0 dB. Selection occurs by ID, not by achieved benefit. Each folder supplies clean, noisy, baseline, selected and oracle WAVs. The comparison sampler plays clean, noisy, baseline and selected with one-second gaps. PCM-24 WAVs preserve the common scale; MP3 samplers are provided for convenient listening. The audio page also shows the reference transcript.

For example 1, the reference is: "YET THESE THOUGHTS AFFECTED HESTER PRYNNE LESS WITH HOPE THAN APPREHENSION". Listen for preserved words and residual interference separately. A selected floor chosen for WER is not a guarantee of the most pleasant sound. Waveform and unit-RMS noise-spectrum figures are included in the supplementary results.

9. Discussion, limitations and reflection

9.1 What the investigation establishes

The selected floor reduces corpus WER by 4.75 points against baseline and 3.13 against noisy input; it improves on baseline for nine of ten speakers. However, the environment split qualifies that average: white-noise WER falls from 43.14% to 32.13%, while traffic rises from 12.36% to 13.36% and cafeteria rises from 29.33% to 29.96%. Thus, the average benefit is driven by the synthetic condition. Calibration-time separation and speaker transfer do not establish a recognition benefit in these two real environments.

The multi-objective result is consistent with literature distinguishing waveform, quality and intelligibility goals [1,3,4]. The selected floor sacrifices 0.68 dB mean SNR relative to baseline while lowering WER. Weak speech and interference both pass through less attenuated bins. This explains the engineering trade-off, but not which phonetic cues cause individual errors. Linguistic priors and decoding can also change transcripts; no model-internal causal explanation is established.

9.2 Noise estimation and the oracle comparison

Oracle PSD has the highest aggregate SNR (7.44 dB), but WER is 30.39%, worse than noisy speech. More accurate access to average noise power is therefore insufficient here. Time averaging still misses local variation; speech-power estimation and noisy phase remain imperfect. A preceding calibration can additionally mismatch the following noise. These measurements separate outcomes, without isolating the causal contribution of each limitation.

9.3 Limits and threats to validity

Coverage: The 20 test clips and ten speakers are selected by ID and duration, not randomly. Only read English is tested. Metadata sex balance does not establish gender, accent or language coverage. Each real environment is represented by one recording; disjoint times do not make them unseen environments.

Parameter selection: Four development utterances make the selection unstable. The fixed grid avoids test tuning but does not establish an optimum. One noise draw per condition also leaves seed sensitivity unmeasured.

Metrics and decoding: STOI predicts intelligibility; it does not replace listeners. WER depends on tiny.en, int8 and the fixed decoder guards. Token caps can truncate hallucinated output. Another recogniser could change the ranking. All errors remain in the dataset; selected audio examples do not establish listener preference.

Uncertainty: Speaker bootstrap retains repeated conditions but cannot represent unseen environments. Short clips create coarse sentence-WER changes. The baseline WER-change interval includes zero, so its average worsening is not a precise population conclusion.

9.4 What would improve the next iteration?

Next, enlarge development speech, vary mixing seeds and evaluate several independent recordings per environment. Compare a time-varying PSD estimator to the static one. Word-aligned inspection, another recogniser and blinded listening would distinguish acoustic preservation, decoder effects and preference.

10. Conclusion and reproducibility

This project implements STFT Wiener filtering and tests a development-selected gain floor on 240 controlled mixtures from 20 utterances and ten unseen test speakers. It includes real traffic/cafeteria interference, separate tuning and test speech, and matched audio examples. Baseline SNR-to-WER disagreements occur in 69 cases. Selected-floor WER is 25.15%, compared with 29.90% for baseline and 28.28% for noisy speech. The contribution is a reproducible task-specific trade-off analysis, not a claim that classical denoising always helps or harms recognition.

Install requirements.txt, then run python src/experiment.py, python src/analyse.py and python src/make_audio.py. The package includes the selected speech/noise files, both measurement CSVs, word transcripts, split manifest, hashes and package/model revisions. Evaluation resumes from condition-level records; use a fresh results directory for an independent full rerun. The brief video demonstrates the saved processing outputs and audio comparisons.

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